

Rosen — §6.4: Binomial Coefficients and Identities

Selected Exercises

Alston

Semester 2, 2026

Exercise 21

Prove that if n and k are integers with $1 \leq k \leq n$, then $k \binom{n}{k} = n \binom{n-1}{k-1}$,

- (a) using a combinatorial proof. [*Hint*: Show that the two sides of the identity count the number of ways to select a subset with k elements from a set with n elements and then an element of this subset.]
- (b) using an algebraic proof based on the formula for $\binom{n}{r}$ given in Theorem 2 in Section 6.3.

Exercise 24

Show that if p is a prime and k is an integer such that $1 \leq k \leq p-1$, then p divides $\binom{p}{k}$.

Exercise 27

Prove the *hockeystick identity*

$$\sum_{k=0}^r \binom{n+k}{k} = \binom{n+r+1}{r}$$

whenever n and r are positive integers,

- (a) using a combinatorial argument.
- (b) using Pascal's identity.

Exercise 31

Show that a nonempty set has the same number of subsets with an odd number of elements as it does subsets with an even number of elements.

Exercise 32

Prove the binomial theorem using mathematical induction.

Exercise 33

In this exercise we will count the number of paths in the xy plane between the origin $(0,0)$ and point (m,n) , where m and n are nonnegative integers, such that each path is made up of a series of steps, where each step is a move one unit to the right or a move one unit upward. (No moves to the left or downward are allowed.)

- (a) Show that each path of the type described can be represented by a bit string consisting of m 0s and n 1s, where a 0 represents a move one unit to the right and a 1 represents a move one unit upward.
- (b) Conclude from part (a) that there are $\binom{m+n}{n}$ paths of the desired type.

Exercise 36

Use Exercise 33 to prove Pascal's identity. [*Hint*: Show that a path of the type described in Exercise 33 from $(0, 0)$ to $(n + 1 - k, k)$ passes through either $(n + 1 - k, k - 1)$ or $(n - k, k)$, but not through both.]

Exercise 38

Give a combinatorial proof that if n is a positive integer then

$$\sum_{k=0}^n k^2 \binom{n}{k} = n(n+1)2^{n-2}.$$

[*Hint*: Show that both sides count the ways to select a subset of a set of n elements together with two not necessarily distinct elements from this subset. Furthermore, express the right-hand side as $n(n-1)2^{n-2} + n2^{n-1}$.]